Phase 1 — API Auth & Security Foundations (design)

Field	Value
Date	2026-05-06
Phase	1 of 6 (parent: `docs/todos/Lava_TODOs_001.md`)
Status	Approved (brainstorm 2026-05-06) — pending plan
Owner	Operator + Claude
Predecessor	post-Ktor cleanup chain ending at `lava-api-go-2.0.16`
Successor	Phase 2 (multi-provider streaming search; absorbs alice-bug routing fix)

1. Decomposition reference

The TODO at `docs/todos/Lava_TODOs_001.md` covers 17 work items. They cannot fit one spec. Decomposition agreed during brainstorm:

#	Phase	TODO sections covered
1	**API auth & security foundations** *(this spec)*	"Lava API secure access" + transport mandate (HTTP/3, brotli, HTTPS body); + α hotfix to hide unsupported providers
2	Multi-provider streaming search	"Search request" + "Receiving results" + per-provider routing (closes alice-404) + events transport
3	First-run onboarding wizard	"Client app first run"
4	Sync subsystem expansion	"Sync now button" + "Additional sync options"
5	UI/UX polish bundle	Menu screen + Color themes + About dialog + Credentials screen + result filtering + nav-bar overlap
6		

#	Phase	TODO sections covered
	Testing sweep + distribution + documentation	"Testing" + Firebase distribute + thinker.local boot + diagrams/ manuals + green-light-to-operator

2. Motivation

Today the `lava-api-go` API has no authentication: any LAN host (or, given the appropriate firewall hole, any internet host) can call its endpoints. The Android client embeds no proof-of-origin in its requests; the API cannot tell a Lava client from `curl`. The TODO mandates: (a) UUID-based client allowlist enforced server-side; (b) progressive backoff on invalid credentials; (c) HTTP/3 with brotli compression and TLS-encrypted bodies for every request and response; (d) build-time UUID injection on the Android client with encrypted-at-rest representation in the APK and decrypt-on-use lifetime in memory.

Phase 1 introduces this auth foundation. Every later phase's networked surface (Phase 2's streaming search, Phase 4's sync, Phase 6's distribution gate) consumes Phase 1's transport.

3. Goals

- **G1** Every API request from the Android client carries a per-build UUID, transported via a config-named HTTP header. The API rejects requests without a valid header.
- **G2** The API stores the allowlist as `HMAC-SHA256(UUID, LAVA_AUTH_HMAC_SECRET)` at boot. Plaintext UUIDs are zeroized after the boot-time hashing pass.
- **G3** Invalid headers from a given source IP trigger a fixed-ladder progressive backoff (`2s → 5s → 10s → 30s → 1m → 1h …`), configurable via `.env`.
- **G4** Per-release UUID rotation: each Android release gets a unique UUID. Active and retired releases are separated; retired releases receive `HTTP 426 Upgrade Required` (no backoff).
- **G5** Transport mandates: HTTP/3 (preferred, with HTTP/2 fallback), brotli response compression, TLS 1.3 minimum.
- **G6** Client-side encrypted UUID at L2 layering — AES-GCM keyed by HKDF over `SHA256(signing-cert)[:16] ‖ pepper`. UUID handled as `ByteArray` end-to-end; never as `String`. Decrypted on the OkHttp interceptor thread; zeroized immediately after request emission.
- **G7** Constitutional clause **§6.R — No-Hardcoding Mandate** added to root `CLAUDE.md` + `AGENTS.md` + every submodule's `CLAUDE.md`. Pre-push enforcement via `scripts/check-constitution.sh`.

- **G8** α-hotfix: Onboarding hides providers whose `TrackerDescriptor.apiSupported = false`; existing installs with such a provider selected receive a one-time dialog directing to another provider.

4. Non-goals (out of scope for Phase 1)

- Multi-provider routing / per-provider search endpoints — Phase 2.
- Streaming results (SSE/WS/gRPC) — Phase 2.
- Onboarding wizard redesign — Phase 3.
- Sync subsystem expansion — Phase 4.
- Color themes / About / Credentials redesign — Phase 5.
- `.env` SIGHUP hot-reload — Phase 6 operations concern.
- Per-installation UUIDs (each device its own) — explicitly rejected during brainstorm; the operator's TODO describes per-build UUIDs.
- Native (NDK) client-side decryption (L3) — explicitly deferred; L2 is the chosen layer.
- Reversible-encryption-at-rest of API allowlist (β) or KDF-hardened comparison (γ) — explicitly rejected during brainstorm.

5. Configuration model (§6.R compliance)

Single `.env` at repo root (existing pattern; `.gitignore`'d per §6.H). New variables introduced by Phase 1:

```
# === Auth field identifier (read by both API + Android build) ===
LAVA_AUTH_FIELD_NAME=Lava-Auth                          # the HTTP
header name
                                                         #
(placeholder; the .env decides)

# === Allowlist + rotation (API reads at boot; Android build picks
one entry) ===
LAVA_AUTH_CURRENT_CLIENT_NAME=android-1.2.7-1027       # which
UUID this build embeds
LAVA_AUTH_ACTIVE_CLIENTS=android-1.2.7-1027:UUID-
A,android-1.2.6-1026:UUID-B
LAVA_AUTH_RETIRED_CLIENTS=android-1.2.5-1025:UUID-C

# === API-only ===
LAVA_AUTH_HMAC_SECRET=<32-byte base64>                 # boot-time
secret for hashing
LAVA_AUTH_BACKOFF_STEPS=2s,5s,10s,30s,1m,1h            # ladder;
configurable
LAVA_AUTH_TRUSTED_PROXIES=                             # CIDR
```

```
list, empty for direct
LAVA_AUTH_MIN_SUPPORTED_VERSION_NAME=1.2.6
LAVA_AUTH_MIN_SUPPORTED_VERSION_CODE=1026

# === Android-build-only ===
LAVA_AUTH_OBFUSCATION_PEPPER=<32-byte base64>          # rotated
per release

# === Transport (API runtime) ===
LAVA_API_HTTP3_ENABLED=true
LAVA_API_BROTLI_QUALITY=4
LAVA_API_BROTLI_RESPONSE_ENABLED=true
LAVA_API_BROTLI_REQUEST_DECODE_ENABLED=false
LAVA_API_PROTOCOL_METRIC_ENABLED=true
```

The same `.env` line `LAVA_AUTH_FIELD_NAME=...` is read by both the API at boot and the Android Gradle codegen at build — single source of truth. The header name is config; nothing in source code names it literally.

`.env.example` (committed) carries placeholder values for every variable above so a developer cloning the repo knows what to set; placeholders match the format but are never valid (e.g. `<32-byte base64>` literal text).

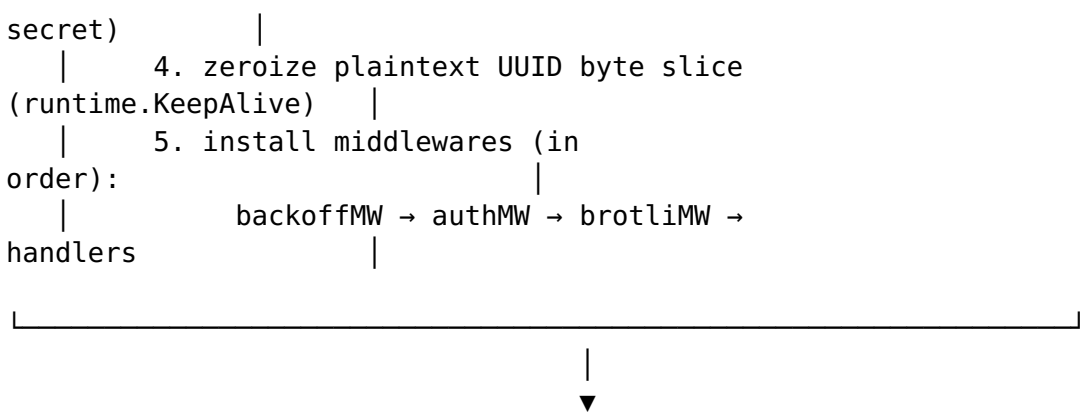
`scripts/check-constitution.sh` extension: greps tracked files for forbidden literal patterns:

- Literal IPv4 addresses outside `.env.example` and incident docs
- The literal header name `Lava-Auth` (or whatever the operator sets — the checker reads from `.env.example`)
- Any 36-char UUID outside `.env.example`
- Hardcoded host:port pairs

Pre-push hook rejects on match.

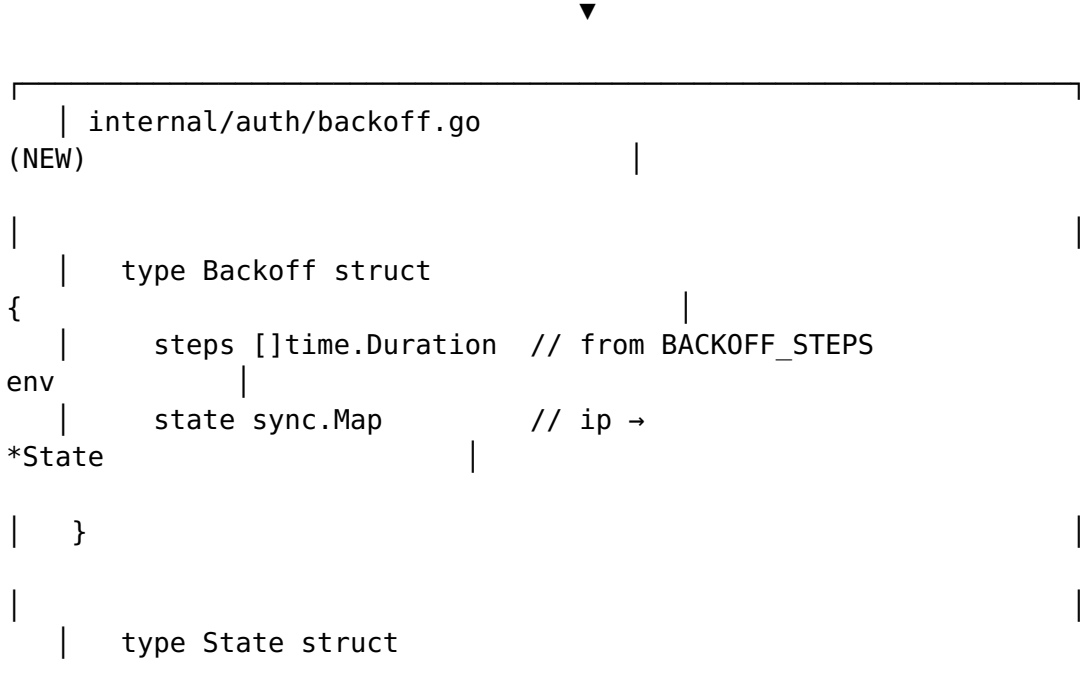
6. API-side architecture

```
┌──────────────────────────────────────────────────────────────────────┐
   │ cmd/lava-api-go/
main.go                                          │
   │
boot:                                                                 │
   │     1. read .env (existing config
layer)                          │
   │     2. parse LAVA_AUTH_ACTIVE_CLIENTS  →
[]ClientEntry         │
   │        parse LAVA_AUTH_RETIRED_CLIENTS →
[]ClientEntry         │
   │     3. for each entry: hash = HMAC-SHA256(uuid,
```


secret)
4. zeroize plaintext UUID byte slice
(runtime.KeepAlive)
5. install middlewares (in
order):
backoffMW → authMW → brotliMW →
handlers


internal/auth/middleware.go
(NEW)
func AuthMiddleware(active, retired
map[string]string,
secret
[]byte,
backoff *Backoff) gin.HandlerFunc
{
return func(c *gin.Context)
{
hdr :=
c.GetHeader(LAVA_AUTH_FIELD_NAME)
if hdr == ""
{ c.AbortWithStatus(401); ... }
blob, err :=
base64.StdEncoding.DecodeString(hdr)
if err != nil
{ c.AbortWithStatus(401); ... }
hash := hmacSHA256(blob,
secret)
if name, ok := active[hash]; ok
{
c.Set("client_name",
name)
backoff.Reset(c.ClientIP())
c.Next()
return
}
if name, ok := retired[hash]; ok
{
c.Header("Lava-Min-Version-
Name",


config.MinSupportedVersionName)
c.JSON(426,
gin.H{
"error":
"client_version_unsupported",
"min_supported_version_name":
config.MinSupportedVersionName,
"min_supported_version_code":
config.MinSupportedVersionCode,
"client_name":
name,
})
c.Abort()
return
}
backoff.RecordFailure(c.ClientIP())
c.AbortWithStatusJSON(401, gin.H{"error":
"unauthorized"})
}
}


internal/auth/backoff.go
(NEW)
type Backoff struct
{
steps []time.Duration // from BACKOFF_STEPS
env
state sync.Map // ip →
*State
}
type State struct

```
{                                                  |
   |     mu
sync.Mutex                                         |
   |     failures
int                                                 |
   |     blockedUntil
time.Time                                          |

|   }                                                                  |

|                                                                      |
   |   func (b *Backoff) CheckBlocked(ip string) (bool,
int)        |
   |     // returns (blocked,
retryAfterSeconds)                     |

|                                                                      |
   |   func (b *Backoff) RecordFailure(ip
string)                     |
   |     // advances ladder; clamps at last
step                      |

|                                                                      |
   |   func (b *Backoff) Reset(ip
string)                             |

|                                                                      |
   |   func BackoffMiddleware(b *Backoff)
gin.HandlerFunc            |
   |     // before authMW: returns 429 + Retry-After when
blocked   |

└──────────────────────────────────────────────────────────────────────┘
```

`internal/auth/multiprovider.go` (existing) is untouched by Phase 1; per-provider concerns live in Phase 2.

`submodules/ratelimiter` already exists. Investigation task: does its API expose fixed-ladder per-key state, or only token-bucket? If fixed-ladder is missing, Phase 1 contributes the ladder pattern UPSTREAM first per the Decoupled Reusable Architecture rule, then pins the new submodule hash in Lava. Lava-side `internal/auth/backoff.go` becomes thin glue. (If the submodule already supports the pattern, the Lava-side file is even thinner.)

7. Client-side architecture

7.1 Build-time codegen

A new Gradle task `:app:generateLavaAuthClass` runs before `compileKotlin`:

1. Read `.env` (already loaded by existing build script via `keystoreLoader` pattern).
2. Look up `LAVA_AUTH_CURRENT_CLIENT_NAME`. If not present → fail build with explicit error.
3. Find the matching entry in `LAVA_AUTH_ACTIVE_CLIENTS`. If not present → fail build.
4. Read the UUID bytes (parse the `xxxx-xxxx-...` representation into 16 raw bytes).
5. Compute the per-build encryption key. The signing-cert hash is the SHA256 of the X.509 DER bytes of the certificate embedded in the variant's keystore (`keystores/<variant>.jks`, path resolved by the existing `keystoreLoader` pattern). Build variant selects keystore: debug variant → debug keystore; release variant → release keystore. The runtime `SigningCertProvider` reads the SAME certificate from `PackageManager.getPackageInfo(... GET_SIGNING_CERTIFICATES)` — DER bytes are identical → hash is identical → derived key matches.

   ```
   certBytes       =
   readKeystore(variant).getCertificate(alias).encoded   // DER
   bytes
   signingCertHash = SHA256(certBytes)[:16]
   pepper          = base64-
   decode(LAVA_AUTH_OBFUSCATION_PEPPER)  // 32 bytes
   key             = HKDF-SHA256(salt = signingCertHash, ikm =
   pepper, info = "lava-auth-v1", length = 32)
   ```

6. AES-GCM encrypt: `(blob, nonce, tag) = AES-256-GCM(uuid_bytes, key, random_nonce_12_bytes)`
7. Generate Kotlin source `app/src/main/generated/kotlin/lava/auth/LavaAuth.kt`:

   ```kotlin
   internal object LavaAuth {
       internal val BLOB:    ByteArray = byteArrayOf(/* 16 + tag
           bytes */)
       internal val NONCE:   ByteArray = byteArrayOf(/* 12 bytes
           */)
       internal val PEPPER:  ByteArray = byteArrayOf(/* 32 bytes
           */)
       internal val FIELD_NAME: String = "<from
           LAVA_AUTH_FIELD_NAME>"
   }
   ```

8. Zero the in-memory plaintext UUID buffer used during codegen.

`@Generated` annotation (or moduleName-style comment) for the generator's identity. Generated file is `.gitignore`'d but committed-to-build-artifacts location is `app/build/generated/lava-auth/`.

R8 keep-rules retain `LavaAuth` as `internal` but obfuscate field names (the field names like `BLOB`, `NONCE` are decoded by the only caller — an inline crypto helper — via reflection-free direct reference).

7.2 Runtime decryption + injection

A new `core/network/impl/AuthInterceptor.kt`:

```
internal class AuthInterceptor @Inject constructor(
    private val signingCertProvider: SigningCertProvider,
) : Interceptor {

    override fun intercept(chain: Interceptor.Chain): Response {
        val keyBytes = ByteArray(32)
        val uuidBytes = ByteArray(16)
        try {
            // Compute the same key the build-time codegen used
            val certHash =
        signingCertProvider.sha256().copyOfRange(0, 16)
            HKDF.deriveKey(
                salt = certHash,
                ikm = LavaAuth.PEPPER,
                info = "lava-auth-v1".toByteArray(Charsets.UTF_8),
                output = keyBytes,
            )
            // AES-GCM decrypt
            AesGcm.decrypt(
                ciphertextWithTag = LavaAuth.BLOB,
                key = keyBytes,
                nonce = LavaAuth.NONCE,
                aad = null,
                output = uuidBytes,
            )
            val headerValue =
        Base64.getEncoder().encodeToString(uuidBytes)

            val req = chain.request().newBuilder()
                .header(LavaAuth.FIELD_NAME, headerValue)
                .build()
            return chain.proceed(req)
        } finally {
            keyBytes.fill(0)
            uuidBytes.fill(0)
```

```
            // headerValue is a String — JVM-immutable, can't be
        zeroed.
            // Acceptable: it leaves Lava code as soon as OkHttp
        consumes it.
            // Mitigation: do NOT log this header anywhere.
        }
    }
}
```

`SigningCertProvider` (also new) returns the `SHA-256` of the running APK's signing certificate via `PackageManager.getPackageInfo(... PackageManager.GET_SIGNING_CERTIFICATES)`. Same cert hash that the build-time codegen used → same derived key → AES-GCM decrypt succeeds. A re-signed APK has a different cert hash → derived key differs → decrypt fails → no header sent → API responds 401. This kills the re-sign-and-redistribute attack vector.

7.3 Memory-hygiene clause (`core/CLAUDE.md`)

Append to `core/CLAUDE.md`:

> **Auth UUID memory hygiene.** The decrypted UUID inside the client MUST be held only as `ByteArray`, never as `String`. The decrypt-use-zeroize lifetime is bounded by a single OkHttp `Interceptor.intercept()` call; the `ByteArray` is `fill(0)`'d in a `finally` block before the function returns. The Base64-encoded header VALUE is a `String` (immutable) but never logged, never persisted, never assigned to a class field, and leaves Lava code as soon as OkHttp consumes it. Adding logging that includes the header value is a constitutional violation; pre-push grep enforces.

8. Transport policy

- API binds two listeners on the same logical port (8443): UDP via `quic-go` for HTTP/3, TCP via `net/http` for HTTP/2.
- Both listeners share one TLS config (`MinVersion: tls.VersionTLS13`).
- Every HTTP/2 response carries `Alt-Svc: h3=":<port>"; ma=86400` so HTTP-3-capable clients upgrade.
- Brotli response middleware applied after `authMW` (no point compressing 401/426 envelopes).
- `Content-Encoding: br` advertised when client `Accept-Encoding` includes `br`.
- Brotli **request** decoding off by default (`LAVA_API_BROTLI_REQUEST_DECODE_ENABLED=false`); request bodies in Phase 1 are JSON `{auth: ..., query: ...}` shapes whose compression overhead exceeds savings. Configurable for forward compatibility.

- New Prometheus counter: `lava_api_request_protocol_total{protocol="h3"|"h2",status="2xx"|"4xx"|"5xx"}`. Visible in the existing observability profile.

9. Rotation workflow

The runbook lives at `docs/RELEASE-ROTATION.md` (created in Phase 1):

1. `uuidgen` → new UUID for the upcoming release.
2. `openssl rand -base64 32` → new pepper.
3. Update `.env`:
   - Append `android-${NEW_VERSION}-${NEW_CODE}:${NEW_UUID}` to `LAVA_AUTH_ACTIVE_CLIENTS`.
   - Set `LAVA_AUTH_CURRENT_CLIENT_NAME=android-${NEW_VERSION}-${NEW_CODE}`.
   - Replace `LAVA_AUTH_OBFUSCATION_PEPPER` with the new value.
4. Build APK via `./build_and_release.sh` (uses the new pepper + new UUID).
5. Distribute via Firebase: `scripts/firebase-distribute.sh`.
6. Restart API: `scripts/distribute-api-remote.sh` (deploys new `.env` to thinker.local + restarts container).
7. Wait until ≥95% of installs (per Crashlytics version distribution) are on the new version.
8. Move the oldest entry from `LAVA_AUTH_ACTIVE_CLIENTS` to `LAVA_AUTH_RETIRED_CLIENTS`.
9. Restart API. Old APKs now receive `426 Upgrade Required` → user upgrades.

If an active UUID leaks before step 7, skip to step 8 immediately for the leaked release. Affected users get `426`, see upgrade dialog, install newer release.

10. α-hotfix: hide unsupported providers

Add `apiSupported: Boolean` to `core/tracker/api/.../TrackerDescriptor` (or wherever the descriptor sealed-interface lives — to be confirmed during plan phase). Initial values:

- `rutracker.org` → `true`
- `rutor.info` → `true`
- `internet-archive` → `false` (flips to `true` in Phase 2)
- Any other registered tracker → audited per descriptor

`feature/onboarding`'s provider list filters `descriptors.where { it.apiSupported }`. Settings → Trackers also marks unsupported descriptors with a "Not yet supported via this backend" subtitle.

Existing installs with an unsupported provider already selected: `MainActivity.onCreate()` checks the persisted selected providers; if any has `apiSupported = false`, surface a one-time `AlertDialog`: "*`is not currently supported via this backend. Please configure another provider.`*" On dismiss, navigate to `Settings → Trackers`. The "one-time" tracking is via a `Preferences` boolean flag.

This is the Phase-1 stopgap. Phase 2 makes IA `apiSupported = true` by adding the routes, and the hotfix becomes inert.

11. Constitutional addition: §6.R

Add the following to root `CLAUDE.md` after §6.Q (so the chain reads … `§6.O, §6.P, §6.Q, §6.R, §6.L`):

> ### §6.R — No-Hardcoding Mandate (added 2026-05-06, FOURTEENTH §6.L invocation)
>
> No connection address, port, header field name, credential, key, salt, secret, schedule, algorithm parameter, or domain literal shall appear as a string/int constant in tracked source code (`.kt`, `.java`, `.go`, `.gradle`, `.kts`, `.xml`, `.yaml`, `.yml`, `.json`, `.sh`). Every such value MUST come from a config source: `.env` (gitignored), generated config class (build-time codegen reading `.env`), runtime env var, or mounted file.
>
> The placeholder file `.env.example` (committed) carries dummy values for every variable so a developer cloning the repo knows what to set.
>
> `scripts/check-constitution.sh` MUST grep tracked files for forbidden literal patterns:
>
> - Any IPv4 address outside `.env.example` and incident docs
> - The header name from `.env.example`'s `LAVA_AUTH_FIELD_NAME`
> - Any 36-char UUID outside `.env.example`
> - Hardcoded `host:port` pairs in HTTP/HTTPS URLs
>
> Pre-push rejects on match. Bluff-Audit stamp required on any commit that adds new config-driven values, demonstrating the no-hardcoding contract test fails when a literal is reintroduced.
>
> Inheritance: applies recursively to every submodule and every new artifact. Submodule constitutions MAY add stricter rules (e.g. "this submodule MAY NOT read environment variables at all; all config arrives via constructor injection") but MUST NOT relax this clause.

Also append to `AGENTS.md` and every submodule's `CLAUDE.md` per §6.F inheritance.

12. Acceptance criteria

Phase 1 is complete when ALL of:

12.1 API-side

- [ ] `internal/auth/middleware.go` exists, exports `AuthMiddleware`. Active-UUID path returns 200; retired-UUID path returns 426 with min-version JSON; unknown-UUID path returns 401.
- [ ] `internal/auth/backoff.go` exists, exports `Backoff` + `BackoffMiddleware`. Per-IP fixed ladder; resets on success.
- [ ] `cmd/lava-api-go/main.go` wires both middlewares before all v1 routes.
- [ ] `.env.example` carries placeholders for every new variable.
- [ ] `internal/config/config.go` reads + validates every new variable; failure to parse is a fatal boot error (no silent defaults).

12.2 Client-side

- [ ] `app/build.gradle.kts` registers `:app:generateLavaAuthClass` task; runs before `compileKotlin`; fails the build if `.env` lookup fails.
- [ ] `app/src/main/generated/kotlin/lava/auth/LavaAuth.kt` is generated (gitignored; checked into build artifacts only).
- [ ] `core/network/impl/AuthInterceptor.kt` exists, registered into the OkHttp client chain via `NetworkModule`.
- [ ] `SigningCertProvider` exists, returns the running APK's signing-cert SHA256.
- [ ] R8 keep-rules retain the `LavaAuth` object's bytes but not its symbol names.

12.3 Transport

- [ ]

- [ ] HTTP/3 listener active on the configured port; `Alt-Svc` header set on every HTTP/2 response.
- [ ] Brotli response middleware compresses for clients sending `Accept-Encoding: br`.
- [ ] `lava_api_request_protocol_total{protocol="h3"|"h2"}` metric increments correctly under both transports.

12.4 Rotation

- [ ] `docs/RELEASE-ROTATION.md` exists with the operator runbook.
- [ ] `scripts/check-constitution.sh` rejects pushes that add a literal UUID outside `.env.example`.
- [ ] `scripts/firebase-distribute.sh` (extended) reads `LAVA_AUTH_CURRENT_CLIENT_NAME` and refuses to distribute if the chosen UUID is not in `LAVA_AUTH_ACTIVE_CLIENTS`.

12.5 α-hotfix

- [ ] `TrackerDescriptor.apiSupported: Boolean` added; existing onboarding consumes it.
- [ ] First-time-only dialog surfaced for installs that already have an unsupported provider selected.
- [ ] Internet Archive descriptor: `apiSupported = false` in Phase 1; flips in Phase 2.

12.6 Constitutional

- [ ] §6.R added to root `CLAUDE.md`, `AGENTS.md`, every `submodules/*/CLAUDE.md`.
- [ ] `scripts/check-constitution.sh` extended for forbidden-literal greps; pre-push hook updated to call it.
- [ ] Falsifiability rehearsal in commit body: deliberate literal reintroduction fails the checker.

12.7 Tests (§6.G real-stack + §6.A contract + §6.N rehearsals)

Test class	Path	Asserts on
`AuthActiveUuidReturns200Test`	`lava-api-go/tests/integration/`	real Gin engine + real Postgres + real HTTP → 200 + correct respon body
`AuthRetiredUuidReturns426Test`	same	retired UUID → 426 + body containing `min_supported_version_ _code`
`AuthUnknownUuidReturns401Test`	same	unknown UUID → 401 counter increments
`AuthBackoffLadderTest`	same	7 sequential 401s; asse response carries `Retry- <BACKOFF_STEPS[i]>`
`AuthBackoffResetsOnSuccessTest`	same	5 fails, 1 success, 1 fail counter is at step 1 not
`AuthBrotliResponseTest`	same	`Accept-Encoding: br` → r carries `Content-Encodin` decompressed body ma uncompressed control
`AuthH3AltSvcAdvertisementTest`	same	HTTP/2 response carri `Svc: h3=":<port>"`
`AuthProtocolMetricTest`	same	metric `lava_api_request_protoc` increments correctly p protocol used
`AuthHeaderInjectionContractTest`	`lava-api-go/tests/contract/`	§6.A contract: `LAVA_AUTH_FIELD_NAME` fr `.env.example` matches w `internal/auth/middlewar` reads (parallels `healthcheck_contract_te` pattern)
`AuthClientFieldNameMatchesApiTest`	Android side, `core/network/impl/src/test/.../`	the build-time-generate `LavaAuth.FIELD_NAME` ma what the API expects (checks `.env.example`)
`C13_AuthHeaderInjectionTest`	`app/src/androidTest/.../challenges/`	Compose UI Challenge §6.G clause 4: real scre real ViewModel → real + AuthInterceptor → re api-go in podman → ass user-visible state ("sea results loaded successf

Test class	Path	Asserts on
`C14_AuthRotationForceUpgradeDialogTest`	same	Compose UI Challenge returns 426 → upgrade renders + user-visible matches expected

For each test class added or modified, a Bluff-Audit stamp goes in the commit body per §6.N. At least one mutation per class must produce a crisp failure message.

12.8 Distribution gate (§6.I + §6.K + §6.P)

- [ ] API binary built via `submodules/containers` per §6.K; binary digest recorded in `.lava-ci-evidence/<tag>/api-build-digest.txt`.
- [ ] Distributed to thinker.local via `scripts/distribute-api-remote.sh`.
- [ ] APK built, signed, distributed via Firebase (`Lava-Android-1.2.7-1027`); §6.P CHANGELOG entry + per-version snapshot at `.lava-ci-evidence/distribute-changelog/firebase/1.2.7-1027.md`.
- [ ] §6.I emulator matrix gate green: phone + tablet on API 28/30/34/latest, all rows pass (`gating: true`, `concurrent: 1`, no AVD-shadow).
- [ ] Operator notified to install + smoke-test the four flows: login, search-against-rutracker, browse, download.

13. Risks + mitigations

Risk	Mitigation
submodules/ratelimiter doesn't expose fixed-ladder per-key state	Plan-phase investigation; if missing, contribute upstream first per Decoupled Reusable Architecture rule (a multi-day extension before Phase 1's Lava-side work begins)
Re-signed-APK auth bypass through derived-key change	Killed by design: signing cert SHA enters HKDF salt; re-sign breaks decrypt
Per-release pepper rotation	`scripts/firebase-distribute.sh` extension: refuse to distribute if `LAVA_AUTH_OBFUSCATION_PEPPER` matches the

Risk	Mitigation
forgotten by operator	previous distribute's value (compares against `.lava-ci-evidence/distribute-changelog/firebase/last-pepper.sha256`)
Generated `LavaAuth.kt` accidentally committed	Add to `.gitignore`; pre-push hook scans for the file path
Header name leaked via logging	The constitutional `core/CLAUDE.md` clause + `scripts/check-constitution.sh` grep for the field name in non-test logging code
HTTP/3 disabled by accident on thinker.local	Boot-time validation: if `LAVA_API_HTTP3_ENABLED=true` but UDP listener fails, abort startup with explicit error (no silent fallback to HTTP/2-only)
Backoff state grows unbounded under brute force	`sync.Map` entries with `blockedUntil + 24h < now` are pruned by a background goroutine

14. Plan-phase investigations

Items the implementation plan (writing-plans) must resolve before phase work begins:

1. **`submodules/ratelimiter` capabilities audit.** Does it already expose fixed-ladder per-key backoff? Is the API stable? If yes, Phase 1's `internal/auth/backoff.go` is thin glue. If no, plan a small upstream contribution to `submodules/ratelimiter` that adds the ladder primitive (with its own §6.A contract test + §6.N falsifiability rehearsal), pin a new submodule hash in Lava, then proceed with Phase 1's Lava-side work. Per Decoupled Reusable Architecture rule: upstream first, Lava pin second.
2. **`TrackerDescriptor.apiSupported` location.** The descriptor sealed-interface lives somewhere under `submodules/tracker_sdk` or `core/tracker`. Plan must locate the canonical site and decide whether `apiSupported` is a constructor parameter or a derived property.
3. **OkHttp `Interceptor` registration site.** Trace the existing `NetworkModule` Hilt graph to confirm where `AuthInterceptor` is added; ensure it runs after the existing `NetworkLogger` (so logger never sees an unredacted header) and before TLS.
4. **Build variant + keystore wiring.** Confirm `keystoreLoader` pattern resolves debug/release keystores correctly; confirm Gradle task can invoke it before `compileKotlin`.

5. **`scripts/check-constitution.sh` regex set.** Plan must enumerate the exact forbidden-literal regexes for §6.R enforcement; ensure pre-push performance is acceptable on the full `git ls-files` scan.
6. **`scripts/firebase-distribute.sh` extension scope.** Confirm where the previous-pepper-SHA is read/written; ensure the same-pepper-rejection check fires reliably.

15. References

- Operator TODO: `docs/todos/Lava_TODOs_001.md`
- Constitution root: `CLAUDE.md` §6.A through §6.Q
- Existing API auth surfaces: `lava-api-go/internal/auth/passthrough.go`, `multiprovider.go`
- Existing client network: `core/network/impl/src/main/kotlin/lava/network/impl/`
- Memory anchor: `feedback_lan_tls_no_manual_install.md` (LAN TLS without manual cert install — must be preserved)
- §6.R inheritance target: every `submodules/*/CLAUDE.md`
- §6.L invocation count: this spec corresponds to the FOURTEENTH operator restatement of the Anti-Bluff Functional Reality Mandate (the no-hardcoding directive piggy-backs on §6.L's spirit and motivates §6.R's mechanics)